

General Relativity HW-5

PROBLEM 1. Consider a weak-field approximation such that $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$. Show that a change

$$h_{\mu\nu} \rightarrow h_{\mu\nu} + 2\epsilon\partial_{(\mu}\xi_{\nu)}$$

gives a physically equivalent space-time for any vector ξ_ν as long as the change is small. In particular, show that the Riemann tensor remains unchanged.

SOLUTION:

By using the result in homework 4, we have the following first order expansion of Riemann tensor:

$$\delta R^\mu_{\alpha\nu\beta} = D_\nu|_{\text{unperturbed}}\gamma^\mu_{\alpha\beta} - D_\beta|_{\text{unperturbed}}\gamma^\mu_{\alpha\nu}$$

And $\gamma^\mu_{\alpha\beta} = \frac{1}{2}(D_\alpha\delta h^\mu_\beta + D_\beta\delta h^\mu_\alpha - D^\mu\delta h_{\alpha\beta})$ Plugging in $\delta h_{\mu\nu} = 2\epsilon\partial_{(\mu}\xi_{\nu)}$ and by using the fact that because we are expanding unperturbed covariant derivative $D_\mu = \partial_\mu$ (you will notice that the perturbation by $h + \delta h$ is identical to the sum of the perturbation caused respectively by h and δh , thus is reasonable to just calculate the perturbation of δh around η , which is the same as the perturbation around g)

By using xact in Mathematica, we have derived that

$$\delta R^\mu_{\alpha\nu\beta} = 0$$

□

The source code can be verified in annex Hw5.nb. The code was run in Wolfram Mathematica 13.0 with package xact 1.2.0.

PROBLEM 2. Show that the harmonic gauge

$$\square x^\mu = 0$$

is equivalent to a condition

$$\partial_\lambda(\sqrt{-g}g^{\lambda\mu}) = 0$$

SOLUTION:

There is some subtlety here because x^μ is actually a scalar field on the manifold (even though it has an upper index μ), because local transformation of tangent space doesn't change the value of x^μ . So $D_\beta x^\mu = \partial_\beta x^\mu$. Then the first equation is equivalent to:

$$\square x^\mu = 0 \implies g^{\alpha\beta}D_\alpha(D_\beta x^\mu) = g^{\alpha\beta}(\partial_\alpha\partial_\beta x^\mu - \Gamma^\nu_{\rho\beta}\partial_\nu x^\mu) = -g^{\alpha\beta}\Gamma^\mu_{\alpha\beta} = 0$$

The second equation is equivalent to:

$$\partial_\lambda(\sqrt{-g}g^{\lambda\mu}) = 0 \implies \sqrt{-g}\partial_\lambda g^{\lambda\mu} + \sqrt{-g}\Gamma^\alpha_{\alpha\lambda}g^{\lambda\mu} = 0 \implies \partial_\lambda g^{\lambda\mu} + \Gamma^\alpha_{\alpha\lambda}g^{\lambda\mu} = 0$$

Where we have used the result in hw2 that $\partial_\lambda \sqrt{-g} = \sqrt{-g} \Gamma_{\alpha\lambda}^\alpha$.

Because the connection is metric compatible, we know that:

$$D_\lambda g^{\lambda\mu} = \partial_\lambda g^{\lambda\mu} + \Gamma_{\lambda\alpha}^\lambda g^{\alpha\mu} + \Gamma_{\alpha\lambda}^\mu g^{\lambda\alpha} = 0$$

Thus

$$\partial_\lambda g^{\lambda\mu} + \Gamma_{\alpha\lambda}^\alpha g^{\lambda\mu} = D_\lambda g^{\lambda\mu} - \Gamma_{\alpha\lambda}^\mu g^{\lambda\alpha} = -\Gamma_{\alpha\lambda}^\mu g^{\lambda\alpha} = 0$$

It can then be observed that two equations are equivalent. □

PROBLEM 3. Show that the Gauss-Bonnet term

$$G = R^2 - 4R_{\mu\nu}^2 + R_{\mu\alpha\nu\beta}^2$$

is a total derivative in 4 dimensions and as such does not contribute to equations of motion. In other words, show that

$$\int d^4x \sqrt{-g} G = \int d^4x \partial_\mu j^\mu$$

SOLUTION:

It suffices to prove that $G\sqrt{-g}d^4x$ is a proper 4-form.

Notice that (with astonishing observation) Let

$$\Omega = \sum_{\sigma \in S_4} \left[\frac{1}{2} R^{\sigma(2)}{}_{\sigma(4)jk} \Gamma_{\sigma(3)i}^{\sigma(1)} + \frac{1}{3} \Gamma_{\sigma(3)i}^{\sigma(1)} \Gamma_{\nu j}^{\sigma(2)} \Gamma_{\sigma(4)k}^\nu \right] dx^i \wedge dx^j \wedge dx^k$$

Then $D\Omega = d(\sqrt{-g}\Omega) = G\sqrt{-g}d^4x$. I tried to verify it, but there are too many calculations. I also tried to put it in xact, but it doesn't seem to work...

PROBLEM 4. (Extra problem of last homework) Derive the second variation of the Einstein-Hilbert action around Minkowski space-time in 4 dimensions. The result should be

$$\delta^2 S = \int d^4x \sqrt{-\eta} \frac{M_P^2}{2} \left[\frac{1}{4} h_{\mu\nu} \square h^{\mu\nu} - \frac{1}{4} h \square h + \frac{1}{2} h \partial_\mu \partial_\nu h^{\mu\nu} + \frac{1}{2} \partial_\mu h^{\mu\rho} \partial_\nu h_\rho^\nu \right]$$

SOLUTION:

$$\mathcal{L} = \sqrt{-g} R$$

By results in hw 4, we can expand the action to first order:

$$\delta S = \frac{M_P^2}{2} \int d^4x \sqrt{-g} (\delta R - \frac{1}{2} g_{\mu\nu} h^{\mu\nu} R) = \frac{M_P^2}{2} \int d^4x \sqrt{-g} (-R_{\mu\nu} + D_\mu D_\nu - g_{\mu\nu} \square - \frac{1}{2} g_{\mu\nu} R) h^{\mu\nu}$$

In order to get the second variation, we take variation with respect to the coefficient of $h^{\mu\nu}$:

$$\delta \left[\sqrt{-g} (-R_{\mu\nu} + D_\mu D_\nu - g_{\mu\nu} \square - \frac{1}{2} g_{\mu\nu} R) \right]$$

¹from 流形与几何初步 by 梅加强, P184-188

$$\begin{aligned}
&= \sqrt{-g} \left(-\frac{1}{2} (\partial_\alpha \partial_\mu h_\nu^\alpha + \partial_\alpha \partial_\nu h_\mu^\alpha - \square h_{\mu\nu} - \partial_\mu \partial_\nu h) - h_{\mu\nu} \square - \frac{1}{2} g_{\mu\nu} (\partial_\alpha \partial_\beta - g_{\alpha\beta} \square) h^{\alpha\beta} \right) \\
&\quad - \frac{1}{2} \sqrt{-g} g^{\rho\theta} h_{\rho\theta} (\partial_\mu \partial_\nu - g_{\mu\nu} \square) \\
&\implies \delta \left[\sqrt{-g} (-R_{\mu\nu} + D_\mu D_\nu - g_{\mu\nu} \square - \frac{1}{2} g_{\mu\nu} R) \right] h^{\mu\nu} \\
&= \sqrt{-g} \left(-\frac{1}{2} \partial_\alpha h_\nu^\alpha \partial_\mu h^{\mu\nu} - \frac{1}{2} \partial_\alpha h_\mu^\alpha \partial_\nu h^{\mu\nu} + \frac{1}{2} \partial_\mu \partial_\nu (h h^{\mu\nu}) - \frac{1}{2} h \partial_\mu \partial_\nu h^{\mu\nu} + \frac{1}{2} h \square h - \frac{1}{2} h \partial_\mu \partial_\nu h^{\mu\nu} \right) \\
&= \sqrt{-g} \left(-\partial_\mu h^{\mu\nu} \partial_\alpha h_\nu^\alpha + \frac{1}{2} h \square h - h \partial_\mu \partial_\nu h^{\mu\nu} + \frac{1}{2} h_{\mu\nu} \square h^{\mu\nu} \right)
\end{aligned}$$

(Remark: the term $\frac{1}{2} h_{\mu\nu} \square h^{\mu\nu}$ comes from $\frac{1}{2} \partial_\mu \partial_\nu (h) h^{\mu\nu} = h^{\mu\nu} g^{\rho\theta} \partial_\mu \partial_\nu h_{\rho\theta}$)

Thus by noticing

$$\delta^2 S = \frac{1}{2} \left(\frac{M_p^2}{2} \int d^4 x \delta \left[\sqrt{-g} (-R_{\mu\nu} + D_\mu D_\nu - g_{\mu\nu} \square - \frac{1}{2} g_{\mu\nu} R) \right] h^{\mu\nu} \right)$$

we obtain the desired result. (well there seems to exist a sign problem...)

□